

Regression Analysis Assignment 1

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INTRODUCTION

This reports consists of detailed statistical analysis using regression model to study two data sets and study the relationship between the variables.

1. Air Quality and Factories – Examining the impact of rise in industries on air quality by analyzing the correlation between the number of factories and the Air Quality Index (AQI).
2. Electricity Consumption and Temperature – Investigating how temperature fluctuations influence electricity usage in households.

METHODOLOGY

The analysis was conducted manually and on R studio both. Calculations were done using MS Excel. Linear Regression Model Fitting

The Least Squares Method was used to compute the slope and the intercept. The regression equation $Y = \beta_0 + \beta_1 X$ was derived for prediction. ANOVA table was constructed and F-test via ANOVA determined whether the independent variable significantly influenced the dependent variable. Hypothesis testing was also done to find is there is a significant relationship or not. Residual Analysis & Model Assumptions were done using the following plots:

1. Residuals vs. Fitted plot checked for linearity.
2. Q-Q plot tested for normality of residuals.
3. Scale-Location plot assessed constant variance (homoscedasticity).
4. Residuals vs. Leverage plot identified influential points.

QUESTION 1

```
rm(list = ls())  
# Load necessary library  
library(ggplot2)
```

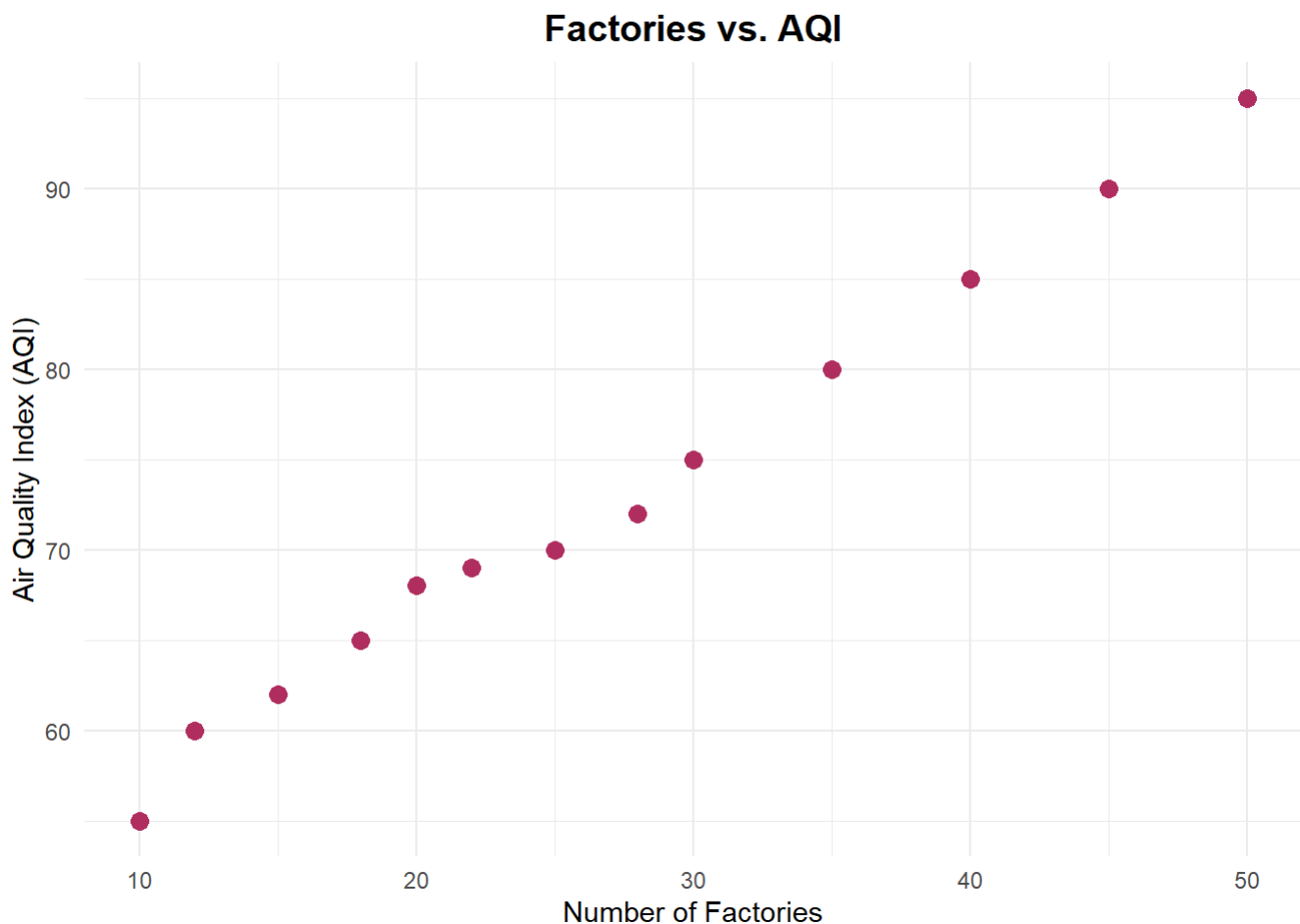
```
## Warning: package 'ggplot2' was built under R version 4.3.3
```

```
# Loading the question 1 data  
factories <- c(10, 25, 30, 20, 12, 35, 50, 28, 15, 18, 40, 22, 45)  
aqi <- c(55, 70, 75, 68, 60, 80, 95, 72, 62, 65, 85, 69, 90)
```

PART A

Scatter plot of the data

```
# Creating the Scatter plot
ggplot(data = data.frame(factories, aqi), aes(x = factories, y = aqi)) +
  geom_point(color = "maroon", size = 3) +
  labs(title = "Factories vs. AQI",
       x = "Number of Factories",
       y = "Air Quality Index (AQI)") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 14, face = "bold"),
  )
```



Inference: The data points on the scatter plot show an upward trend, which suggests a positive relationship between the number of factories and AQI. The relationship seems to be linear, hence a simple linear model might be a good fit for this data.

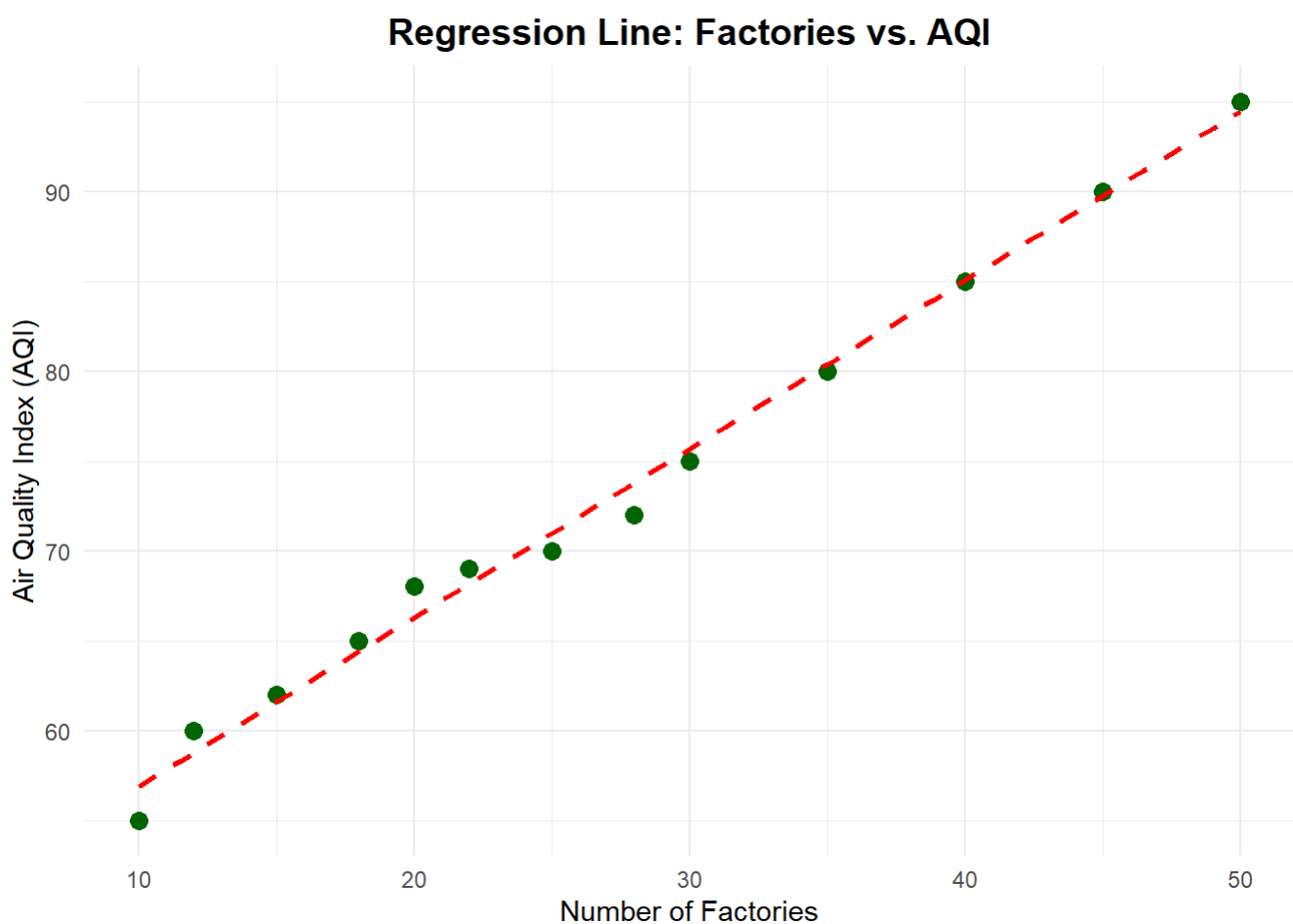
PART B

Superimposing the regression line on the scatterplot

```
# Fit Linear Model
model <- lm(aqi ~ factories)

# Scatterplot with Regression Line
ggplot(data = data.frame(factories, aqi), aes(x = factories, y = aqi)) +
  geom_point(color = "darkgreen", size = 3) +
  geom_smooth(method = "lm", se = FALSE, color = "red", linetype = "dashed") +
  labs(title = "Regression Line: Factories vs. AQI", x = "Number of Factories", y = "Air Quality Index (AQI)") +
  theme_minimal() +
  theme (
    plot.title = element_text(hjust = 0.5, size = 14, face = "bold")
  )
)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Inference: The scatter plot with fitted regression line confirms that there is a strong positive linear relationship between the number of factories and AQI. This indicates that as the number of factories increase the AQI decreases.

PART D

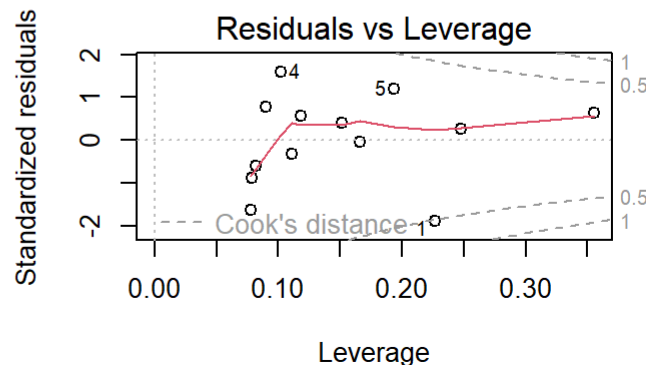
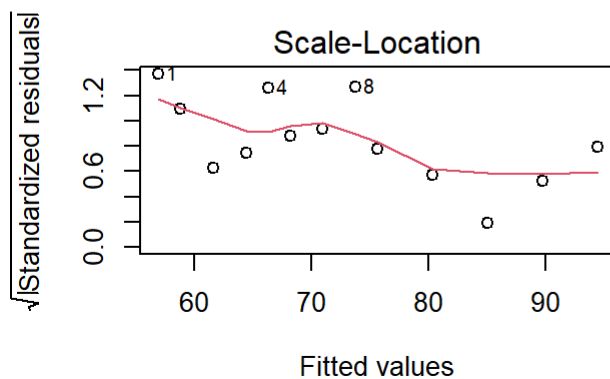
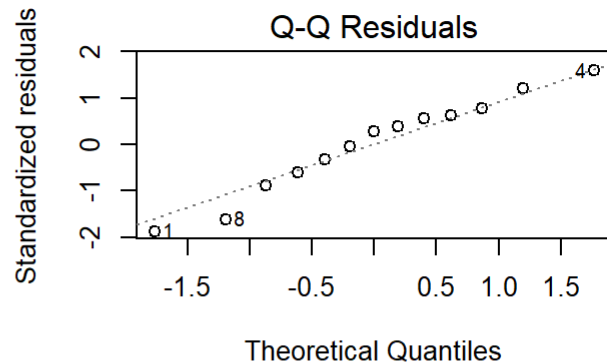
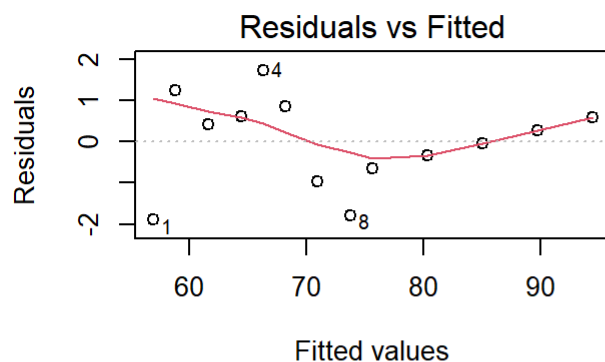
Residual Analysis

```
# Compute residuals
residuals_model <- resid(model)

# Print first few residuals
head(residuals_model)
```

```
##           1           2           3           4           5           6
## -1.8914267 -0.9649349 -0.6561042  1.7262345  1.2321055 -0.3472736
```

```
par(mfrow = c(2, 2))
plot(model)
```



Inference:

Residuals vs Fitted

This plot shows non-linearity. The residuals seem to be randomly scattered around the horizontal dotted line. There is no clear pattern which suggests that residuals have constant variance.

Q-Q Residuals

Since most of the data points fall on the line although there are some deviations the ends. Hence we can that from the QQ plot the residuals seem to be normally distributed.

Scale-Location Plot

The data points appear to be randomly scattered, the trend line is mostly flat which indicates that the residual variance remains quite consistent across the fitted values.

Residuals vs Leverage

Most of the data points fall within the Cook's distance which indicates that there are no extreme influential points. Some points have higher leverage but they don't appear to have a significant influence on the model.

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QUESTION 3

```
# Define the data
electricity <- c(320, 450, 510, 600, 350, 480, 700, 530, 400, 370,
               650, 550, 680, 460, 520, 610, 330, 420, 710, 560)

temperature <- c(15, 12, 10, 8, 14, 11, 5, 9, 13, 14,
               6, 9, 4, 11, 10, 7, 16, 12, 3, 8)

# Combine into a dataframe
data <- data.frame(electricity, temperature)
head(data)
```

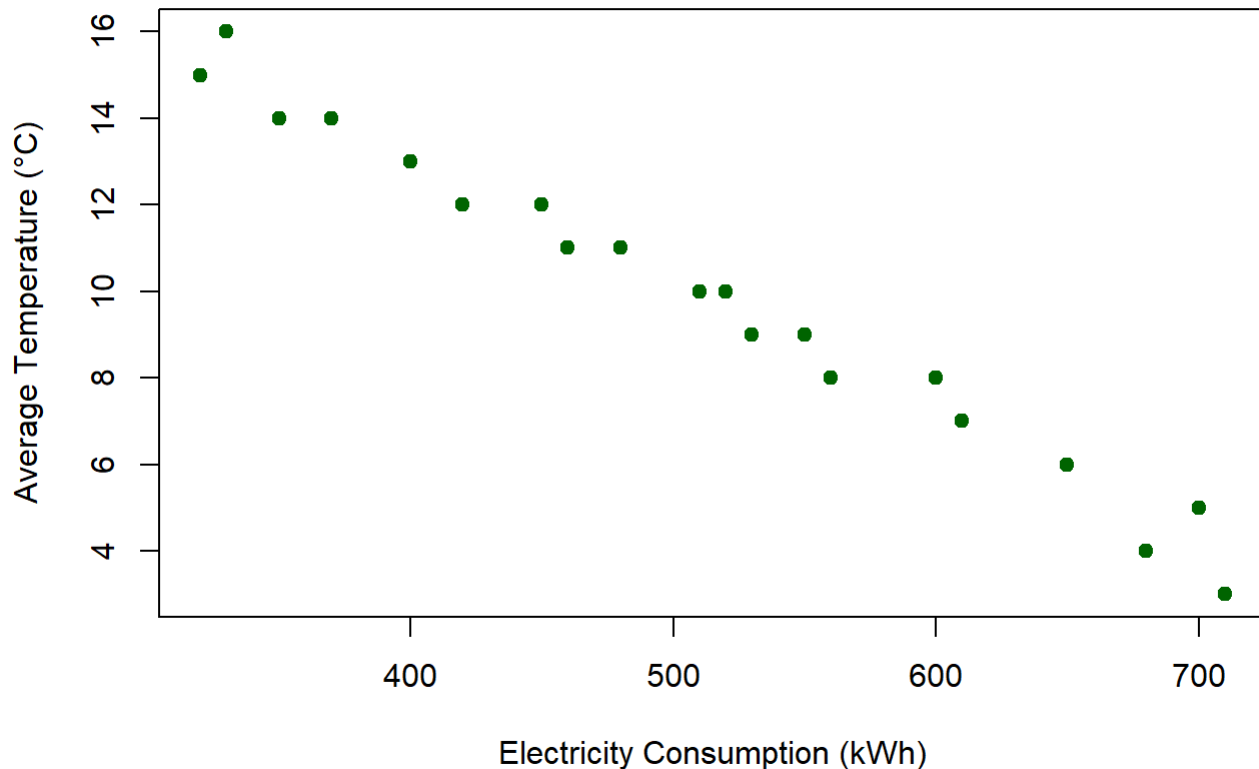
	electricity <dbl>	temperature <dbl>
1	320	15
2	450	12
3	510	10
4	600	8
5	350	14
6	480	11
6 rows		

PART A

Scatter plot of the data

```
# Scatter plot
plot(electricity, temperature,
     main = "Regression Line: Electricity Consumption vs. Temperature",
     xlab = "Electricity Consumption (kWh)",
     ylab = "Average Temperature (°C)",
     col = "darkgreen", pch = 19)
```

Regression Line: Electricity Consumption vs. Temperature

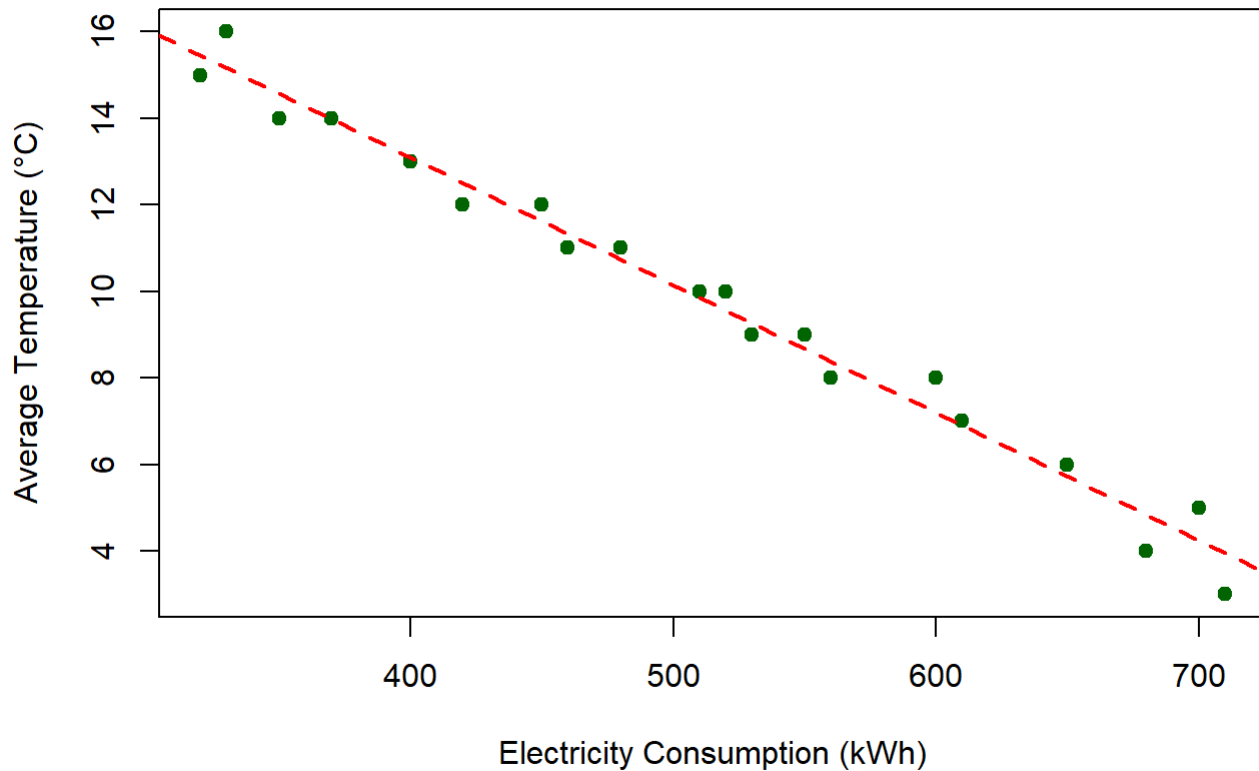


Inference: The scatter plot of electricity data shows a strong negative correlation between electricity consumption and temperature. The data points follow a linear trend which suggests that a linear model might be a good fit.

```
model <- lm(temperature ~ electricity, data = data)
# Superimposing the regression line on the scatter plot
plot(electricity, temperature,
     main = "Regression Line: Electricity Consumption vs. Temperature",
     xlab = "Electricity Consumption (kWh)",
     ylab = "Average Temperature (°C)",
     col = "darkgreen", pch = 19)

# Add Regression Line
abline(model, col = "red", lwd = 2, lty = 2) # Dashed red line
```

Regression Line: Electricity Consumption vs. Temperature



Inference: The scatter plot with fitted regression line confirms that there is a strong negative linear relationship between electricity consumption and temperature. Which means that as the electricity consumption increases the average temperature decreases.

PART B

Fitting the linear regression model to the data

```
# Fit the linear regression model
model <- lm(temperature ~ electricity, data = data)

# Print summary of model
summary(model)
```

```
##
## Call:
## lm(formula = temperature ~ electricity, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.94438 -0.39528  0.05944  0.34292  0.83494
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 24.909334   0.527872   47.19  <2e-16 ***
## electricity -0.029528   0.001008  -29.31  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5409 on 18 degrees of freedom
## Multiple R-squared:  0.9795, Adjusted R-squared:  0.9783
## F-statistic:   859 on 1 and 18 DF,  p-value: < 2.2e-16
```

Inference: The results of the regression analysis show a significant negative relationship between temperature and electricity use. This indicates that with every 0.0295°C drop in temperature for every 1 kWh increase in power use. Temperature is significantly predicted by electricity use, as evidenced by the p-values (<2e-16) for both the intercept and slope. The model explains 97.95% of the temperature change, which is a good fit, with a R square value of 0.9795. Further evidence for the relevance of the model is provided by the F-statistic (859, $p < 2.2e-16$). Furthermore, the residuals are centered around zero, indicating an excellent model fit, and the residual standard error (0.5409) shows little prediction error.

Calculating the residuals for each of the 20 observations.

```
# Compute predicted values
predicted_values <- fitted(model)

# Compute residuals
residuals_values <- residuals(model)

# Create a table with actual, predicted, and residual values
residual_table <- data.frame(
  Electricity_Consumption = electricity,
  Actual_Temperature = temperature,
  Predicted_Temperature = predicted_values,
  Residuals = residuals_values
)

# Display residual table
print(residual_table)
```

##	Electricity_Consumption	Actual_Temperature	Predicted_Temperature	Residuals
## 1	320	15	15.460340	-0.46034004
## 2	450	12	11.621686	0.37831367
## 3	510	10	9.850000	0.15000000
## 4	600	8	7.192471	0.80752949
## 5	350	14	14.574497	-0.57449688
## 6	480	11	10.735843	0.26415684
## 7	700	5	4.239660	0.76034004
## 8	530	9	9.259438	-0.25943789
## 9	400	13	13.098092	-0.09809160
## 10	370	14	13.983935	0.01606523
## 11	650	6	5.716065	0.28393477
## 12	550	9	8.668876	0.33112422
## 13	680	4	4.830222	-0.83022207
## 14	460	11	11.326405	-0.32640527
## 15	520	10	9.554719	0.44528105
## 16	610	7	6.897189	0.10281055
## 17	330	16	15.165059	0.83494101
## 18	420	12	12.507529	-0.50752949
## 19	710	3	3.944379	-0.94437890
## 20	560	8	8.373595	-0.37359473

Inference: High model accuracy is indicated by the table's small residuals, which range from -0.94 to 0.83, and a close match between predicted and actual temperatures. The regression model is valid since the residuals' random distribution indicates that there is no systematic bias. Overall, the model shows excellent predictive performance and little error.

Constructing the ANOVA table

```
# Compute ANOVA Table
anova(model)
```

	Df <int>	Sum Sq <dbl>	Mean Sq <dbl>	F value <dbl>	Pr(>F) <dbl>
electricity	1	251.284178	251.2841777	858.957	1.21052e-16
Residuals	18	5.265822	0.2925457	NA	NA
2 rows					

Inference: The ANOVA table shows a strong model fit and demonstrates that electricity use has significant effects on temperature ($F = 858.96$, $p < 2.2e-16$). The regression's large sum of squares (251.284) compared to its small residual sum (5.266) indicates that electricity consumption accounts for the majority of temperature variation. The statistical significance of this relationship is confirmed by the extremely low p-value.

Finding coefficient of determination and correlation

```
# Compute R-Squared
r_squared <- summary(model)$r.squared

# Compute Correlation Coefficient
correlation <- cor(electricity, temperature)

# Print results
r_squared
```

```
## [1] 0.9794745
```

```
correlation
```

```
## [1] -0.989684
```

Inference: A good model fit is confirmed by the R-squared value (0.9795), which shows that electricity usage influences 97.95% of temperature change. There is a significant relationship between higher power usage and lower temperatures, as indicated by the correlation coefficient (-0.9897). This demonstrates how extremely precise is the regression model.

Finding the standard errors

```
# Extract standard errors from model
std_errors <- summary(model)$coefficients[,2]

# Print standard errors for Intercept & Slope
std_errors
```

```
## (Intercept) electricity
## 0.527872388 0.001007511
```

Inference: While the standard error for the slope (0.0010) demonstrates the accuracy of the electricity consumption coefficient, the standard error for the intercept (0.5279) reveals the uncertainty in determining the baseline temperature. The slope's small standard error indicates that the relationship between temperature and electricity use is extremely reliable.

Hypothesis Testing for significance

Null Hypothesis: There is no relationship between electricity use and temperature.

Alternate Hypothesis: Electricity consumption affects the temperature.

```
# Extract coefficients
beta_coeffs <- summary(model)$coefficients[,1]

# Extract standard errors
std_errors <- summary(model)$coefficients[,2]

# Compute t-values manually
t_values <- beta_coeffs / std_errors

# Print results
t_values
```

```
## (Intercept) electricity
##      47.18817    -29.30797
```

```
# Compute degrees of freedom
df <- nrow(data) - 2 # df = n - 2

# Find critical t-value for two-tailed test at 95% confidence
t_critical <- qt(0.975, df)

# Print critical t-value
t_critical
```

```
## [1] 2.100922
```

Inference:

Since the t-value of slope is greater than the critical t-value, we reject the null hypothesis and state that the slope is statistically significant.

Since the t-value of intercept is greater than the critical t-value, we reject the null hypothesis and state that the intercept is statistically significant as well.

Confidence interval of intercept and slope

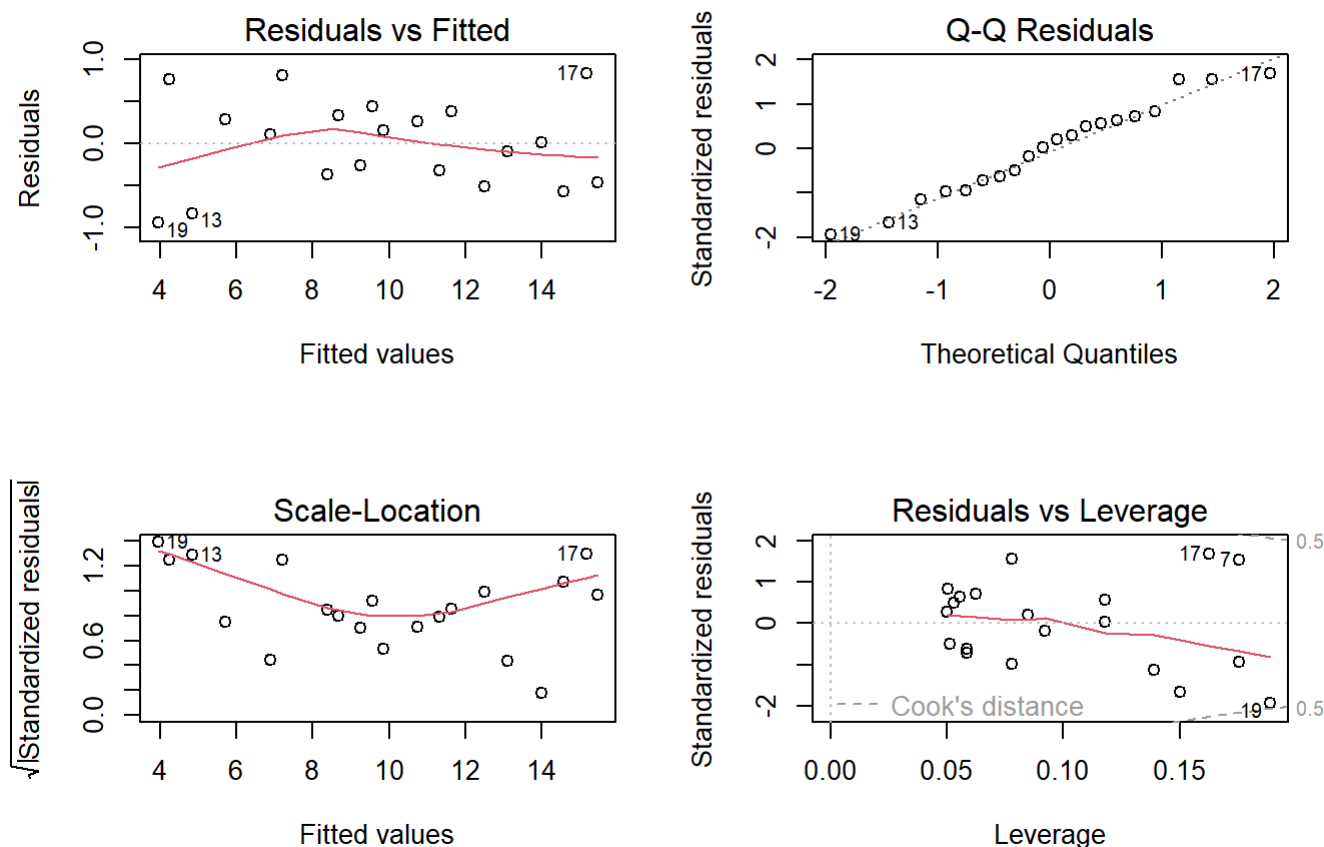
```
# Compute confidence intervals for Intercept & Slope
confint(model, level = 0.95)
```

```
##              2.5 %      97.5 %
## (Intercept) 23.80031506 26.0183525
## electricity -0.03164481 -0.0274114
```

Inference: The shown starting temperature is highly accurate within this range, according to the 95% CI for the intercept (23.80 to 26.02). Since the entire interval stays below zero, the slope's confidence interval (-0.0316 to -0.0274) confirms a steady negative association between temperature and electricity use.

PART C

```
# Generate Residual Analysis Plots
par(mfrow=c(2,2)) # Arrange plots in 2x2 layout
plot(model) # Generates Residuals vs. Fitted, Q-Q, Scale-Location, and Residuals vs. Leverage plots
```



Inference:

Residuals vs Fitted

The plot shows no strong non-linearity, as residuals appear randomly scattered around the horizontal dotted line. There is no clear pattern, suggesting that the variance of residuals remains constant across fitted values.

Q-Q Residuals

Most of the data points align well with the normal line, though there are some slight deviations at the ends. This suggests that residuals are approximately normally distributed.

Scale-Location Plot

The residuals appear randomly spread, and the trend line remains mostly flat, indicating that the variance of residuals remains consistent across fitted values. This confirms that the homoscedasticity assumption holds.

Residuals vs Leverage

Most residuals fall within Cook's distance lines, indicating no extreme influential points. While some observations have higher leverage, they do not significantly impact the regression model.

CONCLUSION

The regression study in this report effectively analysed correlation between number of factories and air quality, and between temperature and electricity use. Results verified a strong positive relationship between factory numbers and Air Quality Index, showing that more factories worsen air quality. The study also found a negative relationship between temperature and electricity use, that is electricity use increases with decrease in temperature. Various statistical methods were used including ANOVA, residual analysis, and confidence intervals which indicated the the models' accuracy. High R-squared values demonstrated that the independent variables substantially explained variations in the dependent variables. Statistical significance was confirmed through F-tests and t-tests.

REFERENCES

1. <https://datatab.net/tutorial/f-distribution> (<https://datatab.net/tutorial/f-distribution>)
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3. <https://rmit.instructure.com/courses/140876/files/44237662?wrap=1>
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4. <https://rmit.instructure.com/courses/140876/files/44224748?wrap=1>
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12. <https://rmit.instructure.com/courses/140876/files/44421022?wrap=1>
(<https://rmit.instructure.com/courses/140876/files/44421022?wrap=1>)
13. <https://rmit.instructure.com/courses/140876/files/44467960?wrap=1>
(<https://rmit.instructure.com/courses/140876/files/44467960?wrap=1>)

USE OF AI

AI was used for improved visualisation of plots using ggplot library. The code given was verified while using it on R console. It was also used to double check my excel calculation. Some concepts which needed more clarification were asked from AI. Everything was double checked manually and from the course modules to ensure accuracy of the result.